

2

Limits and Derivatives

2.2 The Limit of a Function

Let's investigate the behavior of the function f defined by $f(x) = x^2 - x + 2$ for values of x near 2, but not equal to 2.

From the table and the graph of f (a parabola) shown in Figure we see that the closer x is to 2 (on either side of 2), the closer $f(x)$ is to 4. In fact, it appears that we can make the values of $f(x)$ as close as we like to 4 by taking x sufficiently close to 2. We express this by saying “the limit of the function $f(x) = x^2 - x + 2$ as x approaches 2 is equal to 4.” The notation for this is

$$\lim_{x \rightarrow 2} (x^2 - x + 2) = 4$$

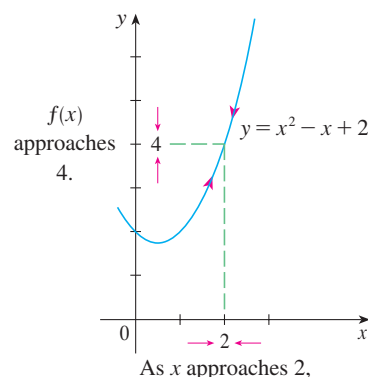
In general, we use the following notation.

1 Intuitive Definition of a Limit Suppose $f(x)$ is defined when x is near the number a . (This means that f is defined on some open interval that contains a , except possibly at a itself.) Then we write

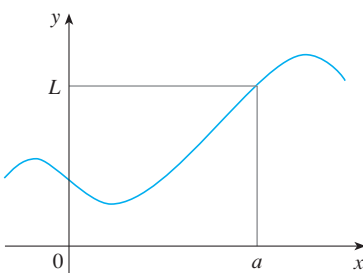
$$\lim_{x \rightarrow a} f(x) = L$$

and say “the limit of $f(x)$, as x approaches a , equals L ”

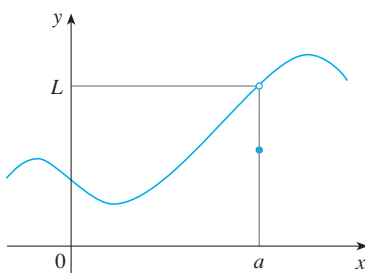
if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by restricting x to be sufficiently close to a (on either side of a) but not equal to a .



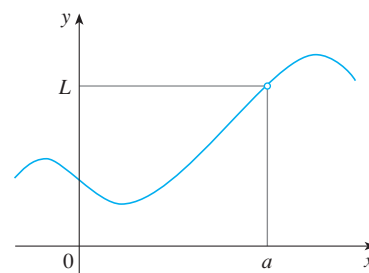
FIGURE



(a)



(b)



(c)

FIGURE $\lim_{x \rightarrow a} f(x) = L$ in all three cases

EXAMPLE 1 Find the value of $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1}$.

SOLUTION

Example 1 is illustrated by the graph of f in Figure 1. Now let's change f slightly by giving it the value 2 when $x = 1$ and calling the resulting function g :

$$g(x) = \begin{cases} \frac{x-1}{x^2-1} & \text{if } x \neq 1 \\ 2 & \text{if } x = 1 \end{cases}$$

This new function g still has the same limit as x approaches 1. (See Figure 2.)

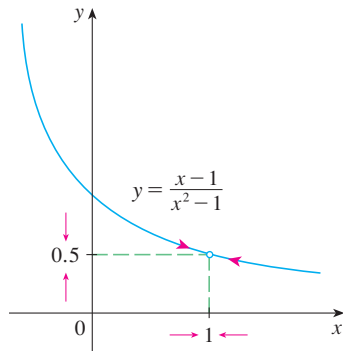


FIGURE 1

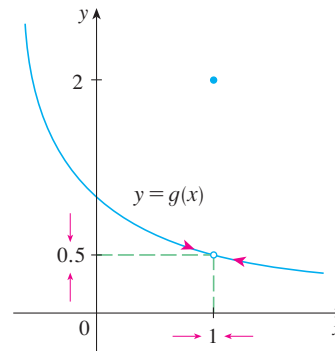


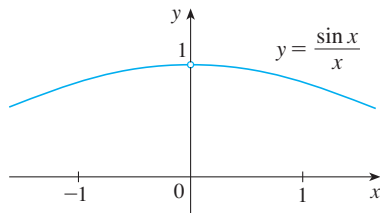
FIGURE 2

EXAMPLE 2 Find the value of $\lim_{t \rightarrow 0} \frac{\sqrt{t^2+9}-3}{t^2}$.

SOLUTION

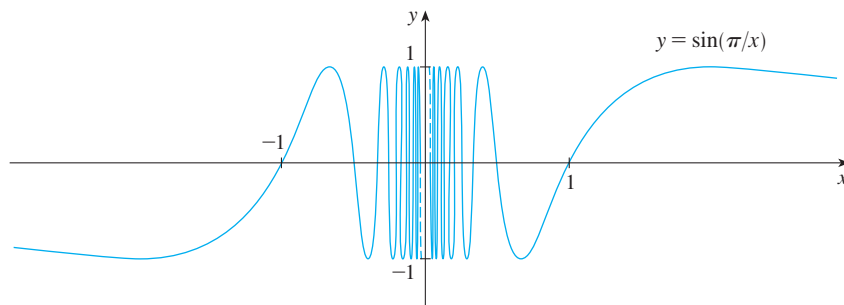
EXAMPLE 3 Find the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

SOLUTION $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$



EXAMPLE 4 Investigate $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$.

SOLUTION $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$ does not exist



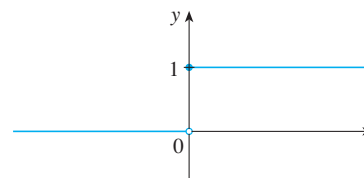
EXAMPLE 5 Find $\lim_{x \rightarrow 0} \left(x^3 + \frac{\cos 5x}{10,000} \right)$.

SOLUTION

EXAMPLE 6 The Heaviside function H is defined by

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

$$\lim_{t \rightarrow 0^-} H(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow 0^+} H(t) = 1$$



The Heaviside function

2 Definition of One-Sided Limits We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the **left-hand limit of $f(x)$ as x approaches a** [or the **limit of $f(x)$ as x approaches a from the left**] is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a with x less than a .

Notice that Definition 2 differs from Definition 1 only in that we require x to be less than a . Similarly, if we require that x be greater than a , we get “the **right-hand limit of $f(x)$ as x approaches a** is equal to L ” and we write

$$\lim_{x \rightarrow a^+} f(x) = L$$

Thus the notation $x \rightarrow a^+$ means that we consider only x greater than a . These definitions are illustrated in Figure 9.

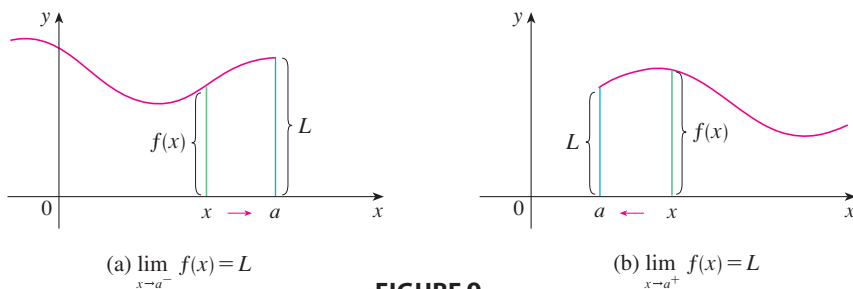


FIGURE 9

3 $\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{x \rightarrow a^-} f(x) = L$ and $\lim_{x \rightarrow a^+} f(x) = L$

EXAMPLE 7 The graph of a function g is shown in Figure. Use it to state the values (if they exist) of the following:

(a) $\lim_{x \rightarrow 2^-} g(x)$

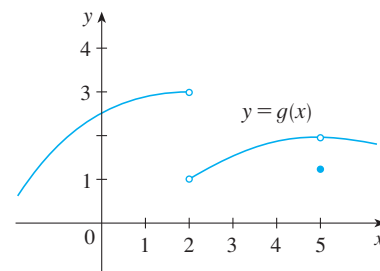
(b) $\lim_{x \rightarrow 2^+} g(x)$

(c) $\lim_{x \rightarrow 2} g(x)$

(d) $\lim_{x \rightarrow 5^-} g(x)$

(e) $\lim_{x \rightarrow 5^+} g(x)$

(f) $\lim_{x \rightarrow 5} g(x)$

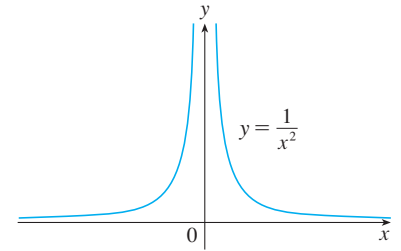


FIGURE

■ Infinite Limits

EXAMPLE 8 Find $\lim_{x \rightarrow 0} \frac{1}{x^2}$ if it exists.

SOLUTION



FIGURE

4 Intuitive Definition of an Infinite Limit Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a , but not equal to a .

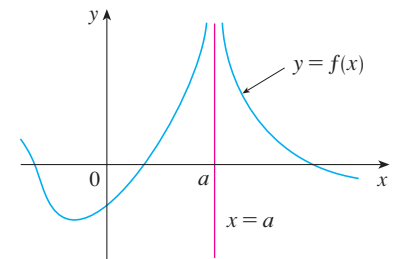
Another notation for $\lim_{x \rightarrow a} f(x) = \infty$ is

$$f(x) \rightarrow \infty \quad \text{as} \quad x \rightarrow a$$

Again, the symbol ∞ is not a number, but the expression $\lim_{x \rightarrow a} f(x) = \infty$ is often read as

“the limit of $f(x)$, as x approaches a , is infinity”

or “ $f(x)$ becomes infinite as x approaches a ”



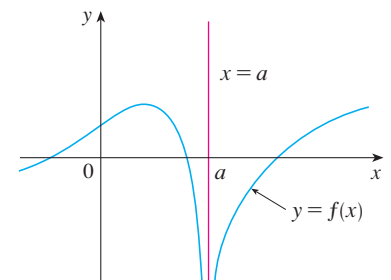
FIGURE

$$\lim_{x \rightarrow a} f(x) = \infty$$

5 Definition Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a , but not equal to a .



FIGURE

$$\lim_{x \rightarrow a} f(x) = -\infty$$

Similar definitions can be given for the one-sided infinite limits

$$\lim_{x \rightarrow a^-} f(x) = \infty \qquad \lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty \qquad \lim_{x \rightarrow a^+} f(x) = -\infty$$

remembering that $x \rightarrow a^-$ means that we consider only values of x that are less than a , and similarly $x \rightarrow a^+$ means that we consider only $x > a$. Illustrations of these four cases are given in Figure

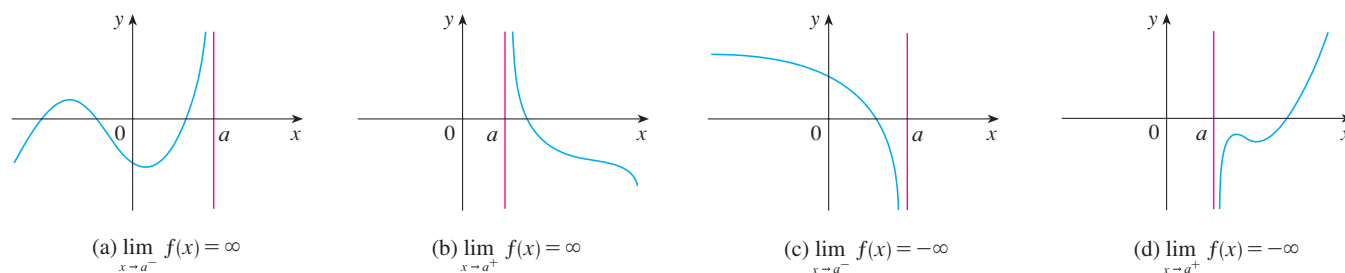


FIGURE 3

6 Definition The vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\begin{array}{lll} \lim_{x \rightarrow a} f(x) = \infty & \lim_{x \rightarrow a^-} f(x) = \infty & \lim_{x \rightarrow a^+} f(x) = \infty \\ \lim_{x \rightarrow a} f(x) = -\infty & \lim_{x \rightarrow a^-} f(x) = -\infty & \lim_{x \rightarrow a^+} f(x) = -\infty \end{array}$$

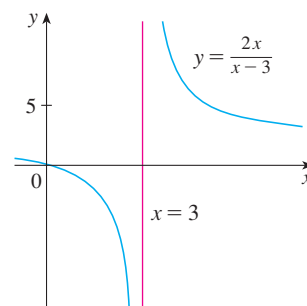
For instance, the y -axis is a vertical asymptote of the curve $y = 1/x^2$ because $\lim_{x \rightarrow 0} (1/x^2) = \infty$. In Figure 3 the line $x = a$ is a vertical asymptote in each of

EXAMPLE 9 Find $\lim_{x \rightarrow 3^+} \frac{2x}{x-3}$ and $\lim_{x \rightarrow 3^-} \frac{2x}{x-3}$.

SOLUTION

$$\lim_{x \rightarrow 3^+} \frac{2x}{x-3} = \infty$$

$$\lim_{x \rightarrow 3^-} \frac{2x}{x-3} = -\infty$$



EXAMPLE 10 Find the vertical asymptotes of $f(x) = \tan x$.

SOLUTION Because

$$\lim_{x \rightarrow (\pi/2)^-} \tan x = \infty \quad \text{and} \quad \lim_{x \rightarrow (\pi/2)^+} \tan x = -\infty$$

This shows that the line $x = \pi/2$ is a vertical asymptote. Similar reasoning shows that the lines $x = \pi/2 + n\pi$, where n is an integer, are all vertical asymptotes of $f(x) = \tan x$. The graph in Figure 4 confirms this. ■

Another example of a function whose graph has a vertical asymptote is the natural logarithmic function $y = \ln x$. From Figure 5 we see that

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

and so the line $x = 0$ (the y -axis) is a vertical asymptote. In fact, the same is true for $y = \log_b x$ provided that $b > 1$. (See Figures 1.5.11 and 1.5.12.)

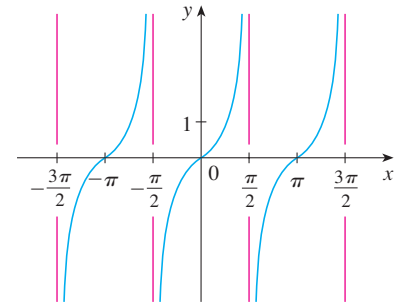


FIGURE 4

$y = \tan x$

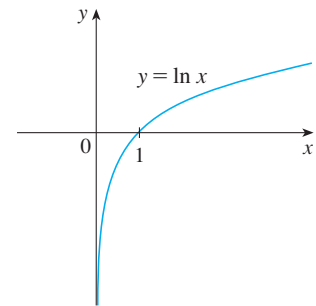


FIGURE 5

The y -axis is a vertical asymptote of the natural logarithmic function.

2.2 EXERCISES

4. Use the given graph of f to state the value of each quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{x \rightarrow 2^-} f(x)$

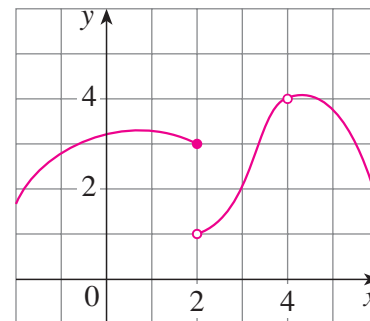
(b) $\lim_{x \rightarrow 2^+} f(x)$

(c) $\lim_{x \rightarrow 2} f(x)$

(d) $f(2)$

(e) $\lim_{x \rightarrow 4} f(x)$

(f) $f(4)$

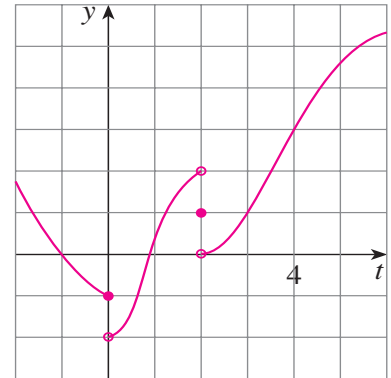


7. For the function g whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{t \rightarrow 0^-} g(t)$ (b) $\lim_{t \rightarrow 0^+} g(t)$ (c) $\lim_{t \rightarrow 0} g(t)$

(d) $\lim_{t \rightarrow 2^-} g(t)$ (e) $\lim_{t \rightarrow 2^+} g(t)$ (f) $\lim_{t \rightarrow 2} g(t)$

(g) $g(2)$ (h) $\lim_{t \rightarrow 4} g(t)$

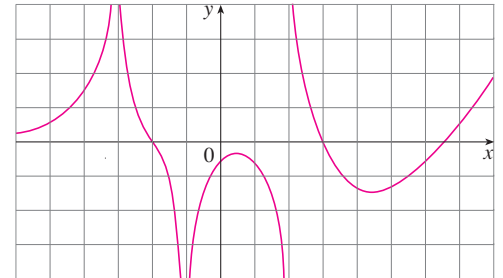


8. For the function A whose graph is shown, state the following.

(a) $\lim_{x \rightarrow -3} A(x)$ (b) $\lim_{x \rightarrow 2^-} A(x)$

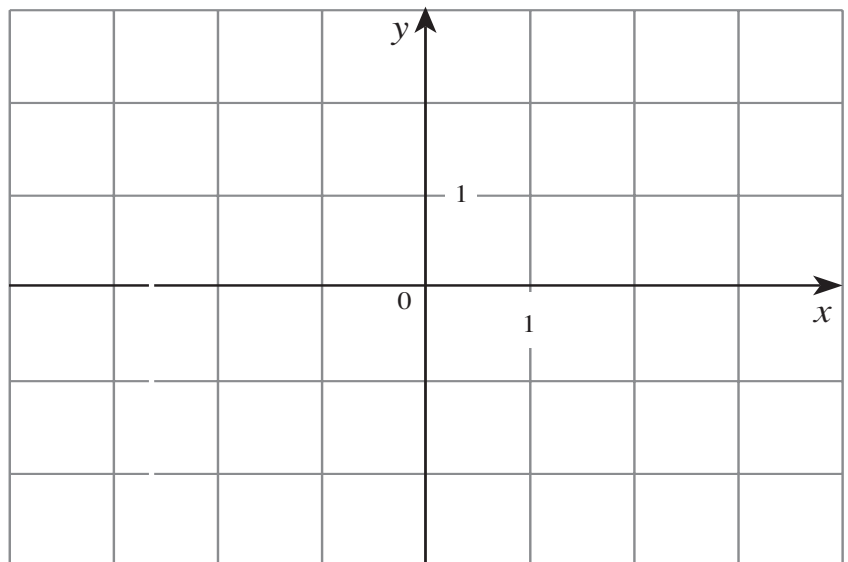
(c) $\lim_{x \rightarrow 2^+} A(x)$ (d) $\lim_{x \rightarrow -1} A(x)$

(e) The equations of the vertical asymptotes



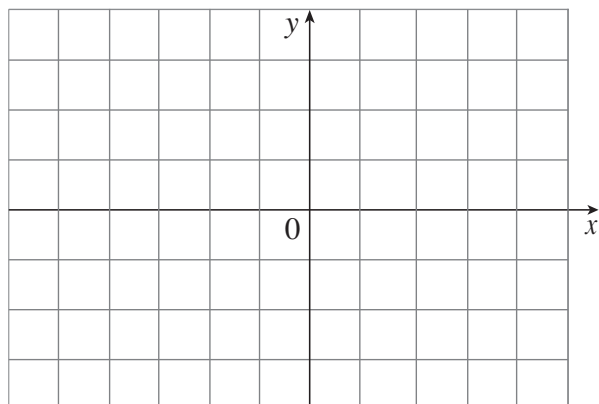
11–12 Sketch the graph of the function and use it to determine the values of a for which $\lim_{x \rightarrow a} f(x)$ exists.

$$f(x) = \begin{cases} 1 + x & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x < 1 \\ 2 - x & \text{if } x \geq 1 \end{cases}$$

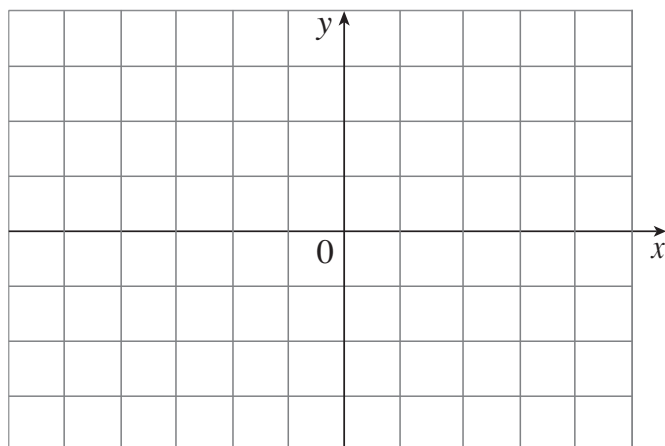


15–18 Sketch the graph of an example of a function f that satisfies all of the given conditions.

16. $\lim_{x \rightarrow 0^-} f(x) = 1$, $\lim_{x \rightarrow 3^-} f(x) = -2$, $\lim_{x \rightarrow 3^+} f(x) = 2$,
 $f(0) = -1$, $f(3) = 1$



18. $\lim_{x \rightarrow 0^-} f(x) = 2$, $\lim_{x \rightarrow 0^+} f(x) = 0$, $\lim_{x \rightarrow 4^-} f(x) = 3$,
 $\lim_{x \rightarrow 4^+} f(x) = 0$, $f(0) = 2$, $f(4) = 1$



31–43 Determine the infinite limit.

31. $\lim_{x \rightarrow 5^+} \frac{x+1}{x-5}$

32. $\lim_{x \rightarrow 5^-} \frac{x+1}{x-5}$

33. $\lim_{x \rightarrow 1} \frac{2-x}{(x-1)^2}$

34. $\lim_{x \rightarrow 3^-} \frac{\sqrt{x}}{(x-3)^5}$

35. $\lim_{x \rightarrow 3^+} \ln(x^2 - 9)$

36. $\lim_{x \rightarrow 0^+} \ln(\sin x)$

39. $\lim_{x \rightarrow 2\pi^-} x \csc x$

40. $\lim_{x \rightarrow 2^-} \frac{x^2 - 2x}{x^2 - 4x + 4}$

41. $\lim_{x \rightarrow 2^+} \frac{x^2 - 2x - 8}{x^2 - 5x + 6}$

42. $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \ln x \right)$

44. Find the vertical asymptotes of the function

$$y = \frac{x^2 + 1}{3x - 2x^2}$$

Limit Laws Suppose that c is a constant and the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist. Then

$$1. \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$2. \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$3. \lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$$

$$4. \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$5. \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

EXAMPLE 1 Use the Limit Laws and the graphs of f and g in Figure 1 to evaluate the following limits, if they exist.

$$(a) \lim_{x \rightarrow -2} [f(x) + 5g(x)] \quad (b) \lim_{x \rightarrow 1} [f(x)g(x)] \quad (c) \lim_{x \rightarrow 2} \frac{f(x)}{g(x)}$$

SOLUTION

(a) From the graphs of f and g we see that

$$\lim_{x \rightarrow -2} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow -2} g(x) = -1$$

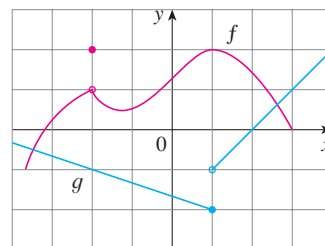


FIGURE 1

$$\begin{aligned} \lim_{x \rightarrow -2} [f(x) + 5g(x)] &= \lim_{x \rightarrow -2} f(x) + \lim_{x \rightarrow -2} [5g(x)] && \text{(by Limit Law 1)} \\ &= \lim_{x \rightarrow -2} f(x) + 5 \lim_{x \rightarrow -2} g(x) && \text{(by Limit Law 3)} \\ &= 1 + 5(-1) = -4 \end{aligned}$$

(b) We see that $\lim_{x \rightarrow 1} f(x) = 2$. But $\lim_{x \rightarrow 1} g(x)$ does not exist because the left and right limits are different:

$$\begin{aligned} \lim_{x \rightarrow 1^-} g(x) &= -2 & \lim_{x \rightarrow 1^+} g(x) &= -1 \\ \lim_{x \rightarrow 1^-} [f(x)g(x)] &= \lim_{x \rightarrow 1^-} f(x) \cdot \lim_{x \rightarrow 1^-} g(x) = 2 \cdot (-2) = -4 \\ \lim_{x \rightarrow 1^+} [f(x)g(x)] &= \lim_{x \rightarrow 1^+} f(x) \cdot \lim_{x \rightarrow 1^+} g(x) = 2 \cdot (-1) = -2 \end{aligned}$$

The left and right limits aren't equal, so $\lim_{x \rightarrow 1} [f(x)g(x)]$ does not exist.

(c) The graphs show that

$$\lim_{x \rightarrow 2} f(x) \approx 1.4 \quad \text{and} \quad \lim_{x \rightarrow 2} g(x) = 0$$

Because the limit of the denominator is 0, we can't use Law 5. The given limit does not exist because the denominator approaches 0 while the numerator approaches a nonzero number. ■

If we use the Product Law repeatedly with $g(x) = f(x)$, we obtain the following law.

$$\mathbf{6.} \quad \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \quad \text{where } n \text{ is a positive integer}$$

In applying these six limit laws, we need to use two special limits:

$$\mathbf{7.} \quad \lim_{x \rightarrow a} c = c$$

$$\mathbf{8.} \quad \lim_{x \rightarrow a} x = a$$

These limits are obvious from an intuitive point of view (state them in words or draw graphs of $y = c$ and $y = x$), but proofs based on the precise definition are requested in the exercises for Section 2.4.

If we now put $f(x) = x$ in Law 6 and use Law 8, we get another useful special limit.

$$\mathbf{9.} \quad \lim_{x \rightarrow a} x^n = a^n \quad \text{where } n \text{ is a positive integer}$$

A similar limit holds for roots as follows. (For square roots the proof is outlined in Exercise 2.4.37.)

$$\mathbf{10.} \quad \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a} \quad \text{where } n \text{ is a positive integer}$$

(If n is even, we assume that $a > 0$.)

More generally, we have the following law, which is proved in Section 2.5 as a consequence of Law 10.

$$\mathbf{11.} \quad \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} \quad \text{where } n \text{ is a positive integer}$$

[If n is even, we assume that $\lim_{x \rightarrow a} f(x) > 0$.]

EXAMPLE 2 Evaluate the following limits and justify each step.

(a) $\lim_{x \rightarrow 5} (2x^2 - 3x + 4)$

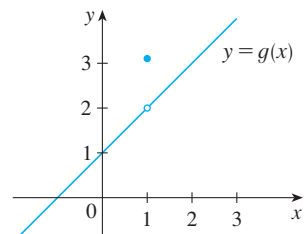
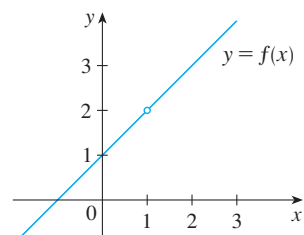
(b) $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$

SOLUTION

Direct Substitution Property If f is a polynomial or a rational function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

EXAMPLE 3 Find $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.



If $f(x) = g(x)$ when $x \neq a$, then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$, provided the limits exist.

EXAMPLE 4 Find $\lim_{x \rightarrow 1} g(x)$ where $g(x) = \begin{cases} x + 1 & \text{if } x \neq 1 \\ \pi & \text{if } x = 1 \end{cases}$

SOLUTION

EXAMPLE 5 Evaluate $\lim_{h \rightarrow 0} \frac{(3 + h)^2 - 9}{h}$.

SOLUTION

EXAMPLE 6 Find $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$.

SOLUTION

1 Theorem $\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$

EXAMPLE 7 Show that $\lim_{x \rightarrow 0} |x| = 0$.

SOLUTION

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

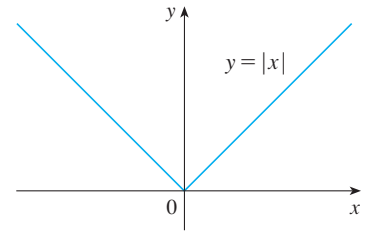


FIGURE 3

EXAMPLE 8 Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

SOLUTION

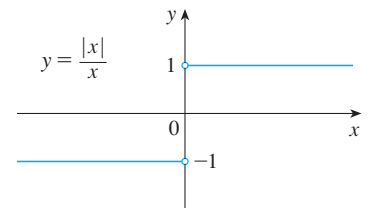


FIGURE 4

EXAMPLE 9 If

$$f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x < 4 \end{cases}$$

determine whether $\lim_{x \rightarrow 4} f(x)$ exists.

SOLUTION

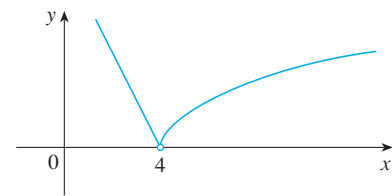


FIGURE 5

EXAMPLE 10 The **greatest integer function** is defined by $\llbracket x \rrbracket =$ the largest integer that is less than or equal to x . (For instance, $\llbracket 4 \rrbracket = 4$, $\llbracket 4.8 \rrbracket = 4$, $\llbracket \pi \rrbracket = 3$, $\llbracket \sqrt{2} \rrbracket = 1$, $\llbracket -\frac{1}{2} \rrbracket = -1$.) Show that $\lim_{x \rightarrow 3} \llbracket x \rrbracket$ does not exist.

SOLUTION The graph of the greatest integer function is shown in Figure 6. Since $\llbracket x \rrbracket = 3$ for $3 \leq x < 4$, we have

$$\lim_{x \rightarrow 3^+} \llbracket x \rrbracket = \lim_{x \rightarrow 3^+} 3 = 3$$

Since $\llbracket x \rrbracket = 2$ for $2 \leq x < 3$, we have

$$\lim_{x \rightarrow 3^-} \llbracket x \rrbracket = \lim_{x \rightarrow 3^-} 2 = 2$$

Because these one-sided limits are not equal, $\lim_{x \rightarrow 3} \llbracket x \rrbracket$ does not exist by Theorem 1. ■

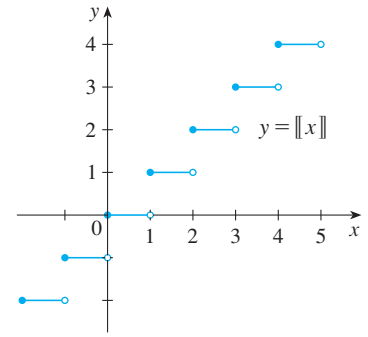


FIGURE 6

Greatest integer function

2 Theorem If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a , then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

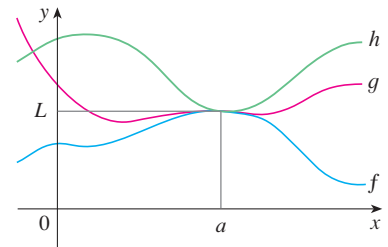


FIGURE 7

3 The Squeeze Theorem If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} g(x) = L$$

EXAMPLE 11 Show that $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$.

SOLUTION First note that we **cannot** use

❌ $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \cdot \lim_{x \rightarrow 0} \sin \frac{1}{x}$

1. Given that

$$\lim_{x \rightarrow 2} f(x) = 4 \quad \lim_{x \rightarrow 2} g(x) = -2 \quad \lim_{x \rightarrow 2} h(x) = 0$$

find the limits that exist. If the limit does not exist, explain why.

(a) $\lim_{x \rightarrow 2} [f(x) + 5g(x)]$

(c) $\lim_{x \rightarrow 2} \sqrt{f(x)}$

(e) $\lim_{x \rightarrow 2} \frac{g(x)}{h(x)}$

11–32 Evaluate the limit, if it exists.

11. $\lim_{x \rightarrow 5} \frac{x^2 - 6x + 5}{x - 5}$

12. $\lim_{x \rightarrow -3} \frac{x^2 + 3x}{x^2 - x - 12}$

13. $\lim_{x \rightarrow 5} \frac{x^2 - 5x + 6}{x - 5}$

14. $\lim_{x \rightarrow 4} \frac{x^2 + 3x}{x^2 - x - 12}$

15. $\lim_{t \rightarrow -3} \frac{t^2 - 9}{2t^2 + 7t + 3}$

16. $\lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{x^2 - 2x - 3}$

17. $\lim_{h \rightarrow 0} \frac{(-5 + h)^2 - 25}{h}$

18. $\lim_{h \rightarrow 0} \frac{(2 + h)^3 - 8}{h}$

19. $\lim_{x \rightarrow -2} \frac{x+2}{x^3+8}$

20. $\lim_{t \rightarrow 1} \frac{t^4-1}{t^3-1}$

25. $\lim_{t \rightarrow 0} \frac{\sqrt{1+t} - \sqrt{1-t}}{t}$

26. $\lim_{t \rightarrow 0} \left(\frac{1}{t} - \frac{1}{t^2+t} \right)$

27. $\lim_{x \rightarrow 16} \frac{4-\sqrt{x}}{16x-x^2}$

29. $\lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{1}{t} \right)$

39. Prove that $\lim_{x \rightarrow 0^+} x^4 \cos \frac{2}{x} = 0$.

40. Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\pi/x)} = 0$.

41–46 Find the limit, if it exists. If the limit does not exist, explain why.

41. $\lim_{x \rightarrow 3} (2x + |x - 3|)$

42. $\lim_{x \rightarrow -6} \frac{2x + 12}{|x + 6|}$

43. $\lim_{x \rightarrow 0.5^-} \frac{2x - 1}{|2x^3 - x^2|}$

44. $\lim_{x \rightarrow -2} \frac{2 - |x|}{2 + x}$

45. $\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

46. $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right)$

51. Let $B(t) = \begin{cases} 4 - \frac{1}{2}t & \text{if } t < 2 \\ \sqrt{t + c} & \text{if } t \geq 2 \end{cases}$ Find the value of c so that $\lim_{t \rightarrow 2} B(t)$ exists.

53. (a) If the symbol $\llbracket \cdot \rrbracket$ denotes the greatest integer function defined in Example 10, evaluate

(i) $\lim_{x \rightarrow -2^+} \llbracket x \rrbracket$

(ii) $\lim_{x \rightarrow -2} \llbracket x \rrbracket$

(iii) $\lim_{x \rightarrow -2.4} \llbracket x \rrbracket$

(b) If n is an integer, evaluate

(i) $\lim_{x \rightarrow n^-} \llbracket x \rrbracket$

(ii) $\lim_{x \rightarrow n^+} \llbracket x \rrbracket$

(c) For what values of a does $\lim_{x \rightarrow a} \llbracket x \rrbracket$ exist?

55. If $f(x) = \llbracket x \rrbracket + \llbracket -x \rrbracket$, show that $\lim_{x \rightarrow 2} f(x)$ exists but is not equal to $f(2)$.

59. If $\lim_{x \rightarrow 1} \frac{f(x) - 8}{x - 1} = 10$, find $\lim_{x \rightarrow 1} f(x)$.

60. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 5$, find the following limits.

(a) $\lim_{x \rightarrow 0} f(x)$

(b) $\lim_{x \rightarrow 0} \frac{f(x)}{x}$

62. Show by means of an example that $\lim_{x \rightarrow a} [f(x) + g(x)]$ may exist even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exists.

63. Show by means of an example that $\lim_{x \rightarrow a} [f(x)g(x)]$ may exist even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exists.

64. Evaluate $\lim_{x \rightarrow 2} \frac{\sqrt{6-x} - 2}{\sqrt{3-x} - 1}$.

-
65. Is there a number a such that $\lim_{x \rightarrow -2} \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$ exists? If so, find the value of a and the value of the limit.

2.5 Continuity

1 Definition A function f is **continuous at a number a** if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Notice that Definition 1 implicitly requires three things if f is continuous at a :

1. $f(a)$ is defined (that is, a is in the domain of f)
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

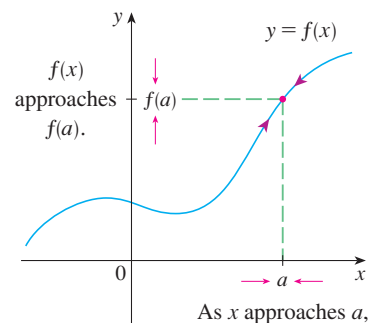


FIGURE 1

EXAMPLE 1 Figure 2 shows the graph of a function f . At which numbers is f discontinuous? Why?

SOLUTION

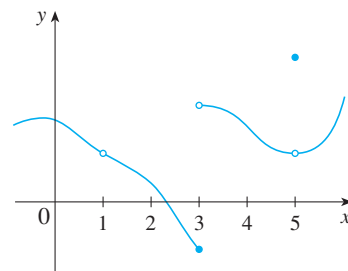


FIGURE 2

EXAMPLE 2 Where are each of the following functions discontinuous?

(a) $f(x) = \frac{x^2 - x - 2}{x - 2}$

(b) $f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$

(c) $f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$

(d) $f(x) = \llbracket x \rrbracket$

SOLUTION

(a) Notice that $f(2)$ is not defined, so f is discontinuous at 2. Later we'll see why f is continuous at all other numbers.

(b) Here $f(0) = 1$ is defined but

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2} \text{ does not exist. So } f \text{ is discontinuous at } 0.$$

(c) Here $f(2) = 1$ is defined and

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3 \text{ exists.}$$

But $\lim_{x \rightarrow 2} f(x) \neq f(2)$

so f is **not continuous** at 2.

(d) The greatest integer function $f(x) = \llbracket x \rrbracket$ has discontinuities at all of the integers because $\lim_{x \rightarrow n} \llbracket x \rrbracket$ does not exist if n is an integer.

Figure 3 shows the graphs of the functions in Example 2. In each case the graph can't be drawn without lifting the pen from the paper because a hole or break or jump occurs in the graph. The kind of discontinuity illustrated in parts (a) and (c) is called **removable** because we could remove the discontinuity by redefining f at just the single number 2. [The function $g(x) = x + 1$ is continuous.] The discontinuity in part (b) is called an **infinite discontinuity**. The discontinuities in part (d) are called **jump discontinuities** because the function “jumps” from one value to another.

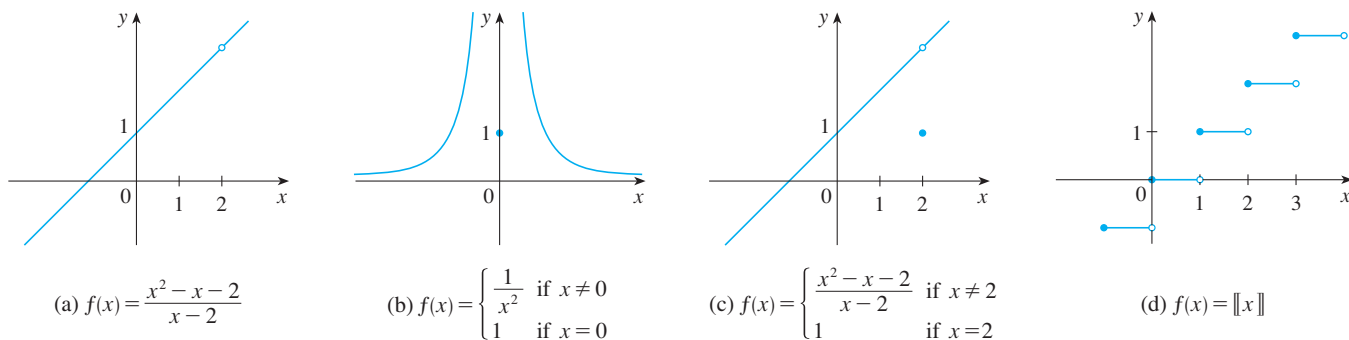


FIGURE 3

Graphs of the functions in Example 2

2 Definition A function f is **continuous from the right at a number a** if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and f is **continuous from the left at a** if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

EXAMPLE 3 At each integer n , the function $f(x) = \llbracket x \rrbracket$ [see Figure 3(d)] is continuous from the right but discontinuous from the left because

$$\lim_{x \rightarrow n^+} f(x) = \lim_{x \rightarrow n^+} \llbracket x \rrbracket = n = f(n)$$

but

$$\lim_{x \rightarrow n^-} f(x) = \lim_{x \rightarrow n^-} \llbracket x \rrbracket = n - 1 \neq f(n)$$



3 Definition A function f is **continuous on an interval** if it is continuous at every number in the interval. (If f is defined only on one side of an endpoint of the interval, we understand *continuous* at the endpoint to mean *continuous from the right* or *continuous from the left*.)

EXAMPLE 4 Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

SOLUTION If $-1 < a < 1$, then using the Limit Laws, we have

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) \\&= 1 - \lim_{x \rightarrow a} \sqrt{1 - x^2} \quad (\text{by Laws 2 and 7}) \\&= 1 - \sqrt{\lim_{x \rightarrow a} (1 - x^2)} \quad (\text{by 11}) \\&= 1 - \sqrt{1 - a^2} \quad (\text{by 2, 7, and 9}) \\&= f(a)\end{aligned}$$

Thus, by Definition 1, f is continuous at a if $-1 < a < 1$. Similar calculations show that

$$\lim_{x \rightarrow -1^+} f(x) = 1 = f(-1) \quad \text{and} \quad \lim_{x \rightarrow 1^-} f(x) = 1 = f(1)$$

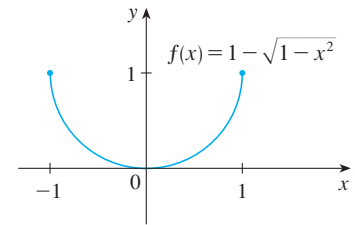


FIGURE 4

4 Theorem If f and g are continuous at a and c is a constant, then the following functions are also continuous at a :

- | | | |
|------------|-----------------------------------|---------|
| 1. $f + g$ | 2. $f - g$ | 3. cf |
| 4. fg | 5. $\frac{f}{g}$ if $g(a) \neq 0$ | |

5 Theorem

- (a) Any polynomial is continuous everywhere; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$.
- (b) Any rational function is continuous wherever it is defined; that is, it is continuous on its domain.

EXAMPLE 5 Find $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$.

SOLUTION

7 Theorem The following types of functions are continuous at every number in their domains:

- polynomials
- rational functions
- root functions
- trigonometric functions
- inverse trigonometric functions
- exponential functions
- logarithmic functions

EXAMPLE 6 Where is the function $f(x) = \frac{\ln x + \tan^{-1}x}{x^2 - 1}$ continuous?

SOLUTION

EXAMPLE 7 Evaluate $\lim_{x \rightarrow \pi} \frac{\sin x}{2 + \cos x}$.

SOLUTION

8 Theorem If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f(b)$.
In other words,

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

EXAMPLE 8 Evaluate $\lim_{x \rightarrow 1} \arcsin\left(\frac{1 - \sqrt{x}}{1 - x}\right)$.

SOLUTION

(a) $h(x) = \sin(x^2)$

(b) $F(x) = \ln(1 + \cos x)$

10 The Intermediate Value Theorem Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

The Intermediate Value Theorem states that a continuous function takes on every intermediate value between the function values $f(a)$ and $f(b)$. It is illustrated by Figure 8. Note that the value N can be taken on once [as in part (a)] or more than once [as in part (b)].

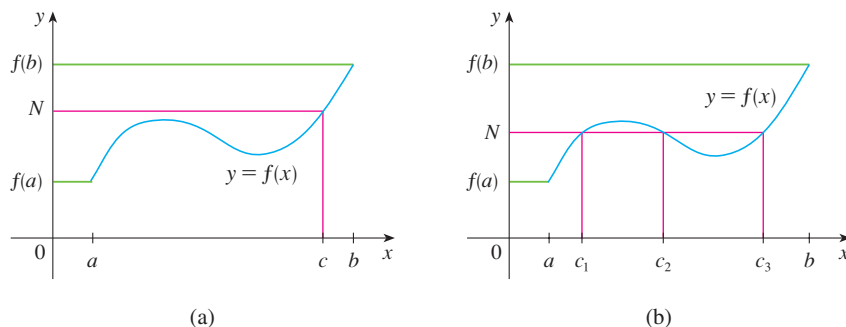
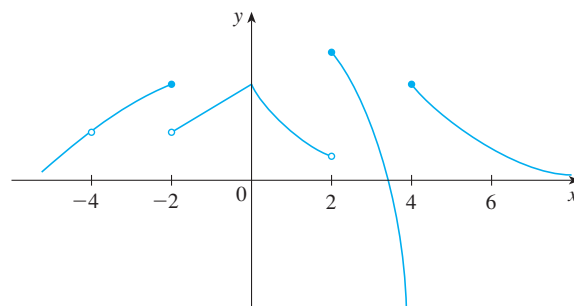


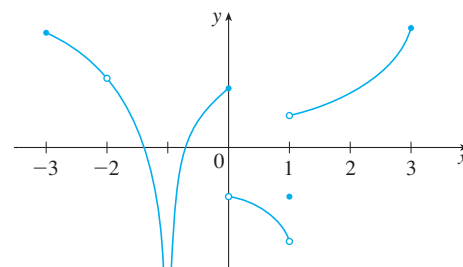
FIGURE 8

EXAMPLE 10 Show that there is a root of the equation $4x^3 - 6x^2 + 3x - 2 = 0$ between 1 and 2.

3. (a) From the graph of f , state the numbers at which f is discontinuous and explain why.
 (b) For each of the numbers stated in part (a), determine whether f is continuous from the right, or from the left, or neither.



4. From the graph of g , state the intervals on which g is continuous.



5–8 Sketch the graph of a function f that is continuous except for the stated discontinuity.

6. Discontinuities at -1 and 4 , but continuous from the left at -1 and from the right at 4

7. Removable discontinuity at 3 , jump discontinuity at 5

8. Neither left nor right continuous at -2 , continuous only from the left at 2

17–22 Explain why the function is discontinuous at the given number a . Sketch the graph of the function.

20. $f(x) = \begin{cases} \frac{x^2 - x}{x^2 - 1} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases} \quad a = 1$

22. $f(x) = \begin{cases} \frac{2x^2 - 5x - 3}{x - 3} & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases} \quad a = 3$

23–24 How would you “remove the discontinuity” of f ? In other words, how would you define $f(2)$ in order to make f continuous at 2?

23. $f(x) = \frac{x^2 - x - 2}{x - 2}$

24. $f(x) = \frac{x^3 - 8}{x^2 - 4}$

39–40 Show that f is continuous on $(-\infty, \infty)$.

39.
$$f(x) = \begin{cases} 1 - x^2 & \text{if } x \leq 1 \\ \ln x & \text{if } x > 1 \end{cases}$$

40.
$$f(x) = \begin{cases} \sin x & \text{if } x < \pi/4 \\ \cos x & \text{if } x \geq \pi/4 \end{cases}$$

45. For what value of the constant c is the function f continuous on $(-\infty, \infty)$?

$$f(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$$

46. Find the values of a and b that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & \text{if } x < 2 \\ ax^2 - bx + 3 & \text{if } 2 \leq x < 3 \\ 2x - a + b & \text{if } x \geq 3 \end{cases}$$

47. Suppose f and g are continuous functions such that $g(2) = 6$ and $\lim_{x \rightarrow 2} [3f(x) + f(x)g(x)] = 36$. Find $f(2)$.

48. Let $f(x) = 1/x$ and $g(x) = 1/x^2$.

(a) Find $(f \circ g)(x)$.

(b) Is $f \circ g$ continuous everywhere? Explain.

49. Which of the following functions f has a removable discontinuity at a ? If the discontinuity is removable, find a function g that agrees with f for $x \neq a$ and is continuous at a .

(a) $f(x) = \frac{x^4 - 1}{x - 1}, \quad a = 1$

(b) $f(x) = \frac{x^3 - x^2 - 2x}{x - 2}, \quad a = 2$

(c) $f(x) = \llbracket \sin x \rrbracket, \quad a = \pi$

51. If $f(x) = x^2 + 10 \sin x$, show that there is a number c such that $f(c) = 1000$.

52. Suppose f is continuous on $[1, 5]$ and the only solutions of the equation $f(x) = 6$ are $x = 1$ and $x = 4$. If $f(2) = 8$, explain why $f(3) > 6$.

53–56 Use the Intermediate Value Theorem to show that there is a root of the given equation in the specified interval.

53. $x^4 + x - 3 = 0$, $(1, 2)$

56. $\sin x = x^2 - x$, $(1, 2)$

58 Prove that the equation $\ln x = 3 - 2x$ has at least one real root.

71. Show that the function

$$f(x) = \begin{cases} x^4 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \text{ is continuous on } (-\infty, \infty).$$

1 Intuitive Definition of a Limit at Infinity Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

2 Definition Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

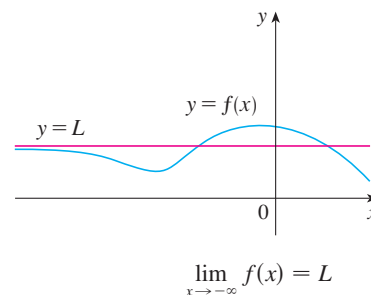
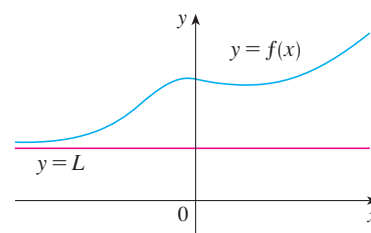
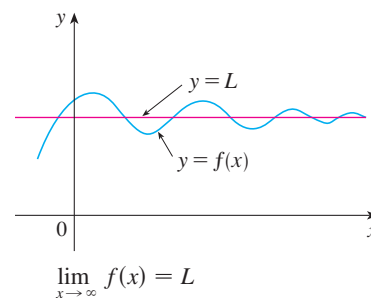
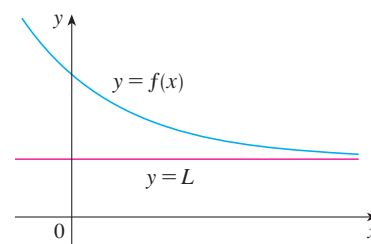
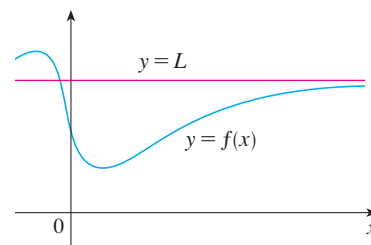
means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large negative.

3 Definition The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

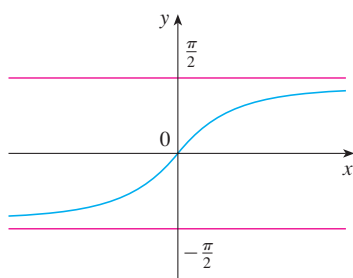
EXAMPLE 1 Find the horizontal asymptotes for the function $f(x) = \frac{x^2 - 1}{x^2 + 1}$

SOLUTION



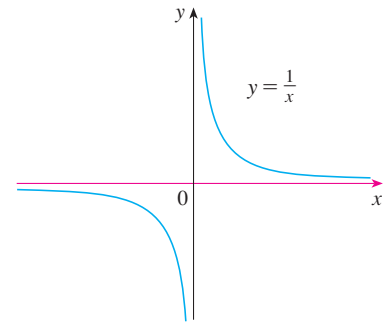
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$$\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2} \quad \lim_{x \rightarrow \infty} \tan^{-1} x = \frac{\pi}{2}$$



EXAMPLE 2 Find $\lim_{x \rightarrow \infty} \frac{1}{x}$ and $\lim_{x \rightarrow -\infty} \frac{1}{x}$.

SOLUTION



$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

5 Theorem If $r > 0$ is a rational number, then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

If $r > 0$ is a rational number such that x^r is defined for all x , then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

EXAMPLE 3 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1}$

SOLUTION

EXAMPLE 4 Find the horizontal and vertical asymptotes of the graph of the function $f(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}$

SOLUTION

EXAMPLE 5 Compute $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x)$.

SOLUTION

EXAMPLE 6 Evaluate $\lim_{x \rightarrow 2^+} \arctan\left(\frac{1}{x-2}\right)$.

SOLUTION

6

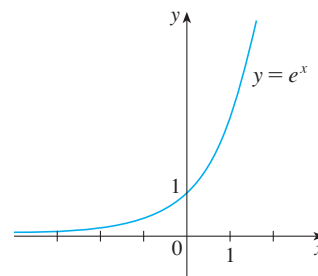
$$\lim_{x \rightarrow -\infty} e^x = 0$$

EXAMPLE 7 Evaluate $\lim_{x \rightarrow 0^-} e^{1/x}$.

SOLUTION If we let $t = 1/x$, we know that $t \rightarrow -\infty$ as $x \rightarrow 0^-$. Therefore, by (6),

$$\lim_{x \rightarrow 0^-} e^{1/x} = \lim_{t \rightarrow -\infty} e^t = 0$$

■



EXAMPLE 8 Evaluate $\lim_{x \rightarrow \infty} \sin x$.

SOLUTION As x increases, the values of $\sin x$ oscillate between 1 and -1 infinitely often and so they don't approach any definite number. Thus $\lim_{x \rightarrow \infty} \sin x$ does not exist. ■

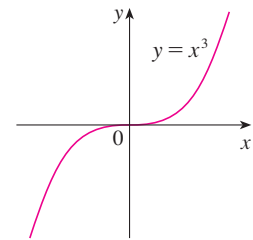
■ Infinite Limits at Infinity

The notation

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

EXAMPLE 9 Find $\lim_{x \rightarrow \infty} x^3$ and $\lim_{x \rightarrow -\infty} x^3$.

SOLUTION



$$\lim_{x \rightarrow \infty} x^3 = \infty, \quad \lim_{x \rightarrow -\infty} x^3 = -\infty$$

EXAMPLE 10 Find $\lim_{x \rightarrow \infty} (x^2 - x)$.

⊗ **SOLUTION** It would be **wrong** to write

$$\lim_{x \rightarrow \infty} (x^2 - x) = \lim_{x \rightarrow \infty} x^2 - \lim_{x \rightarrow \infty} x = \infty - \infty$$

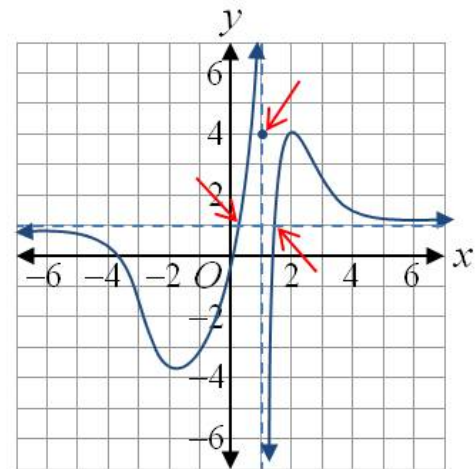
EXAMPLE 11 Find $\lim_{x \rightarrow \infty} \frac{x^2 + x}{3 - x}$.

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SOLUTION

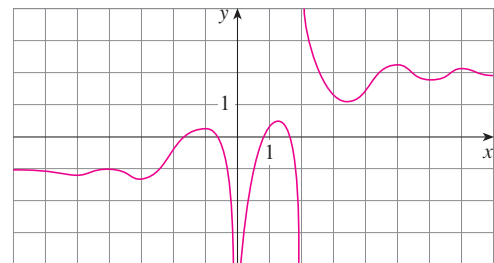
2.6 EXERCISES

2. (a) Can the graph of $y = f(x)$ intersect a vertical asymptote? Can it intersect a horizontal asymptote? Illustrate by sketching graphs.
- (b) How many horizontal asymptotes can the graph of $y = f(x)$ have? Sketch graphs to illustrate the possibilities.



4. For the function g whose graph is given, state the following.

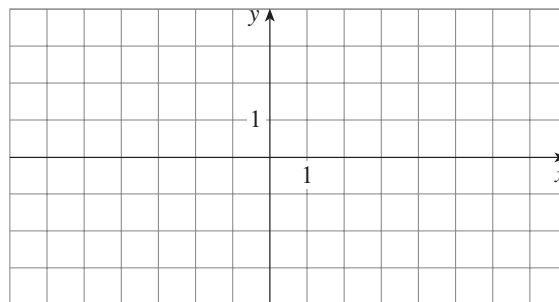
- (a) $\lim_{x \rightarrow \infty} g(x)$ (b) $\lim_{x \rightarrow -\infty} g(x)$
(c) $\lim_{x \rightarrow 0} g(x)$ (d) $\lim_{x \rightarrow 2^-} g(x)$
(e) $\lim_{x \rightarrow 2^+} g(x)$ (f) The equations of the asymptotes



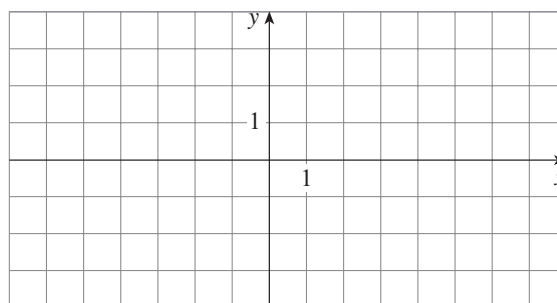
5–10 Sketch the graph of an example of a function f that satisfies all of the given conditions.

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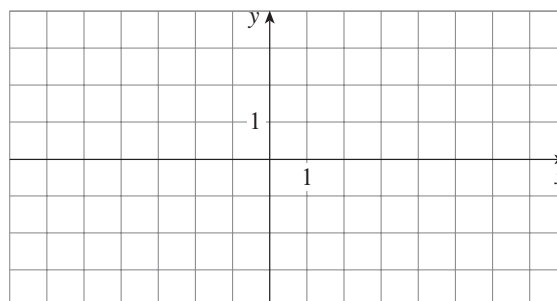
7. $\lim_{x \rightarrow 2} f(x) = -\infty$, $\lim_{x \rightarrow \infty} f(x) = \infty$, $\lim_{x \rightarrow -\infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, $\lim_{x \rightarrow 0^-} f(x) = -\infty$



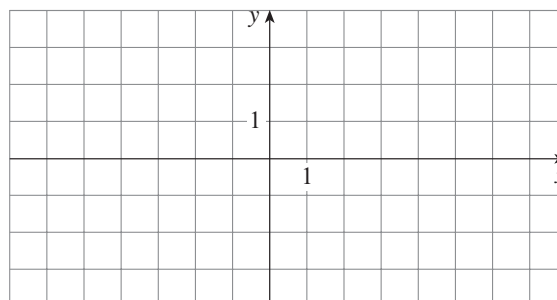
8. $\lim_{x \rightarrow \infty} f(x) = 3$, $\lim_{x \rightarrow 2^-} f(x) = \infty$, $\lim_{x \rightarrow 2^+} f(x) = -\infty$, f is odd



9. $f(0) = 3$, $\lim_{x \rightarrow 0^-} f(x) = 4$, $\lim_{x \rightarrow 0^+} f(x) = 2$, $\lim_{x \rightarrow -\infty} f(x) = -\infty$, $\lim_{x \rightarrow 4^-} f(x) = -\infty$, $\lim_{x \rightarrow 4^+} f(x) = \infty$, $\lim_{x \rightarrow \infty} f(x) = 3$



10. $\lim_{x \rightarrow 3} f(x) = -\infty$, $\lim_{x \rightarrow \infty} f(x) = 2$, $f(0) = 0$, f is even



15–42 Find the limit or show that it does not exist.

18. $\lim_{x \rightarrow -\infty} \frac{4x^3 + 6x^2 - 2}{2x^3 - 4x + 5}$

22. $\lim_{x \rightarrow \infty} \frac{x^2}{\sqrt{x^4 + 1}}$

23. $\lim_{x \rightarrow \infty} \frac{\sqrt{1 + 4x^6}}{2 - x^3}$

24. $\lim_{x \rightarrow -\infty} \frac{\sqrt{1 + 4x^6}}{2 - x^3}$

25. $\lim_{x \rightarrow \infty} \frac{\sqrt{x + 3x^2}}{4x - 1}$

27. $\lim_{x \rightarrow \infty} (\sqrt{9x^2 + x} - 3x)$

28. $\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + 3x} + 2x)$

32. $\lim_{x \rightarrow \infty} (e^{-x} + 2 \cos 3x)$

33. $\lim_{x \rightarrow -\infty} (x^2 + 2x^7)$

35. $\lim_{x \rightarrow \infty} \arctan(e^x)$

37. $\lim_{x \rightarrow \infty} \frac{1 - e^x}{1 + 2e^x}$

38. $\lim_{x \rightarrow \infty} \frac{\sin^2 x}{x^2 + 1}$

39. $\lim_{x \rightarrow \infty} (e^{-2x} \cos x)$

40. $\lim_{x \rightarrow 0^+} \tan^{-1}(\ln x)$

41. $\lim_{x \rightarrow \infty} [\ln(1 + x^2) - \ln(1 + x)]$

42. $\lim_{x \rightarrow \infty} [\ln(2 + x) - \ln(1 + x)]$

43. For $f(x) = \frac{x}{\ln x}$ find each of the following limits.
- (i) $\lim_{x \rightarrow 0^+} f(x)$ (ii) $\lim_{x \rightarrow 1^-} f(x)$ (iii) $\lim_{x \rightarrow 1^+} f(x)$

47–52 Find the horizontal and vertical asymptotes of each curve. If you have a graphing device, check your work by graphing the curve and estimating the asymptotes.

49. $y = \frac{2x^2 + x - 1}{x^2 + x - 2}$

50. $y = \frac{1 + x^4}{x^2 - x^4}$

51. $y = \frac{x^3 - x}{x^2 - 6x + 5}$

52. $y = \frac{2e^x}{e^x - 5}$

58. Find a formula for a function that has vertical asymptotes $x = 1$ and $x = 3$ and horizontal asymptote $y = 1$.

59. A function f is a ratio of quadratic functions and has a vertical asymptote $x = 4$ and just one x -intercept, $x = 1$.

60–64 Find the limits as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. Use this information, together with intercepts, to give a rough sketch of the graph.

60. $y = 2x^3 - x^4$

64. $y = x^2(x^2 - 1)^2(x + 2)$

Tangents

If a curve C has equation $y = f(x)$ and we want to find the tangent line to C at the point $P(a, f(a))$, then we consider a nearby point $Q(x, f(x))$, where $x \neq a$, and compute the slope of the secant line PQ :

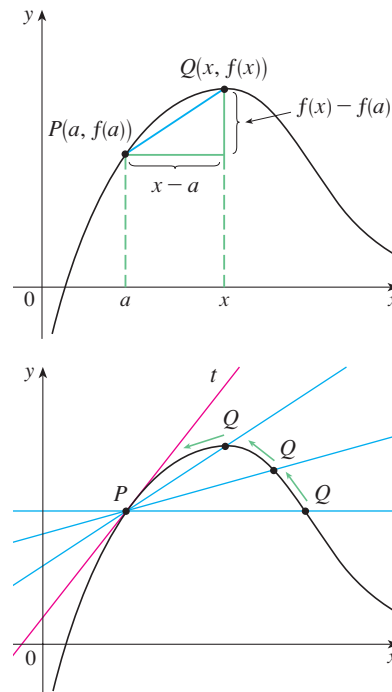
$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$

Then we let Q approach P along the curve C by letting x approach a . If m_{PQ} approaches a number m , then we define the *tangent* t to be the line through P with slope m . (This amounts to saying that the tangent line is the limiting position of the secant line PQ as Q approaches P . See Figure .)

1 Definition The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.



EXAMPLE 1 Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

SOLUTION Here we have $a = 1$ and $f(x) = x^2$, so the slope is

$$\begin{aligned} m &= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \\ &= \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2 \end{aligned}$$

Using the point-slope form of the equation of a line, we find that an equation of the tangent line at $(1, 1)$ is

$$y - 1 = 2(x - 1) \quad \text{or} \quad y = 2x - 1$$

2

$$m = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

EXAMPLE 2 Find an equation of the tangent line to the hyperbola $y = 3/x$ at the point $(3, 1)$.

SOLUTION Let $f(x) = 3/x$. Then, by Equation 2, the slope of the tangent at $(3, 1)$ is

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{3}{3+h} - 1}{h} = \lim_{h \rightarrow 0} \frac{\frac{3 - (3+h)}{3+h}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h(3+h)} = \lim_{h \rightarrow 0} -\frac{1}{3+h} = -\frac{1}{3} \end{aligned}$$

Therefore an equation of the tangent at the point $(3, 1)$ is

$$y - 1 = -\frac{1}{3}(x - 3)$$

which simplifies to $x + 3y - 6 = 0$ ■

4 Definition The **derivative of a function f at a number a** , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

5

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

EXAMPLE 4

Find the derivative of the function $f(x) = x^2 - 8x + 9$ at the number a .

SOLUTION From Definition 4 we have

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(a+h)^2 - 8(a+h) + 9] - [a^2 - 8a + 9]}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^2 + 2ah + h^2 - 8a - 8h + 9 - a^2 + 8a - 9}{h} \\ &= \lim_{h \rightarrow 0} \frac{2ah + h^2 - 8h}{h} = \lim_{h \rightarrow 0} (2a + h - 8) \\ &= 2a - 8 \end{aligned}$$
■

The tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$, the derivative of f at a .

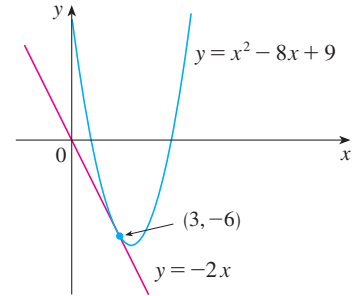
If we use the point-slope form of the equation of a line, we can write an equation of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$:

$$y - f(a) = f'(a)(x - a)$$

EXAMPLE 5 Find an equation of the tangent line to the parabola $y = x^2 - 8x + 9$ at the point $(3, -6)$.

SOLUTION From Example 4 we know that the derivative of $f(x) = x^2 - 8x + 9$ at the number a is $f'(a) = 2a - 8$. Therefore the slope of the tangent line at $(3, -6)$ is $f'(3) = 2(3) - 8 = -2$. Thus an equation of the tangent line, shown in Figure 7, is

$$y - (-6) = (-2)(x - 3) \quad \text{or} \quad y = -2x$$



Rates of Change

Suppose y is a quantity that depends on another quantity x . Thus y is a function of x and we write $y = f(x)$. If x changes from x_1 to x_2 , then the change in x (also called the **increment** of x) is

$$\Delta x = x_2 - x_1$$

and the corresponding change in y is

$$\Delta y = f(x_2) - f(x_1)$$

The difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is called the **average rate of change of y with respect to x** over the interval $[x_1, x_2]$ and can be interpreted as the slope of the secant line PQ .

By analogy with velocity, we consider the average rate of change over smaller and smaller intervals by letting x_2 approach x_1 and therefore letting Δx approach 0. The limit of these average rates of change is called the **(instantaneous) rate of change of y with respect to x** at $x = x_1$, which (as in the case of velocity) is interpreted as the slope of the tangent to the curve $y = f(x)$ at $P(x_1, f(x_1))$:

$$\boxed{6} \quad \text{instantaneous rate of change} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

We recognize this limit as being the derivative $f'(x_1)$.

The derivative $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

9. (a) Find the slope of the tangent to the curve $y = 3 + 4x^2 - 2x^3$ at the point where $x = a$.

(b) Find equations of the tangent lines at the points $(1, 5)$ and $(2, 3)$.

22. If the tangent line to $y = f(x)$ at $(4, 3)$ passes through the point $(0, 2)$, find $f(4)$ and $f'(4)$.

24. Sketch the graph of a function g for which

$$g(0) = g(2) = g(4) = 0, g'(1) = g'(3) = 0,$$

$$g'(0) = g'(4) = 1, g'(2) = -1, \lim_{x \rightarrow \infty} g(x) = \infty, \text{ and}$$

$$\lim_{x \rightarrow -\infty} g(x) = -\infty.$$

26. Sketch the graph of a function f where the domain is $(-2, 2)$,

$$f'(0) = -2, \lim_{x \rightarrow 2^-} f(x) = \infty, f \text{ is continuous at all}$$

numbers in its domain except ± 1 , and f is odd.

39. Find $\lim_{x \rightarrow 2} \frac{x^6 - 64}{x - 2}$

In the preceding section we considered the derivative of a function f at a fixed number a :

$$\boxed{1} \quad f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

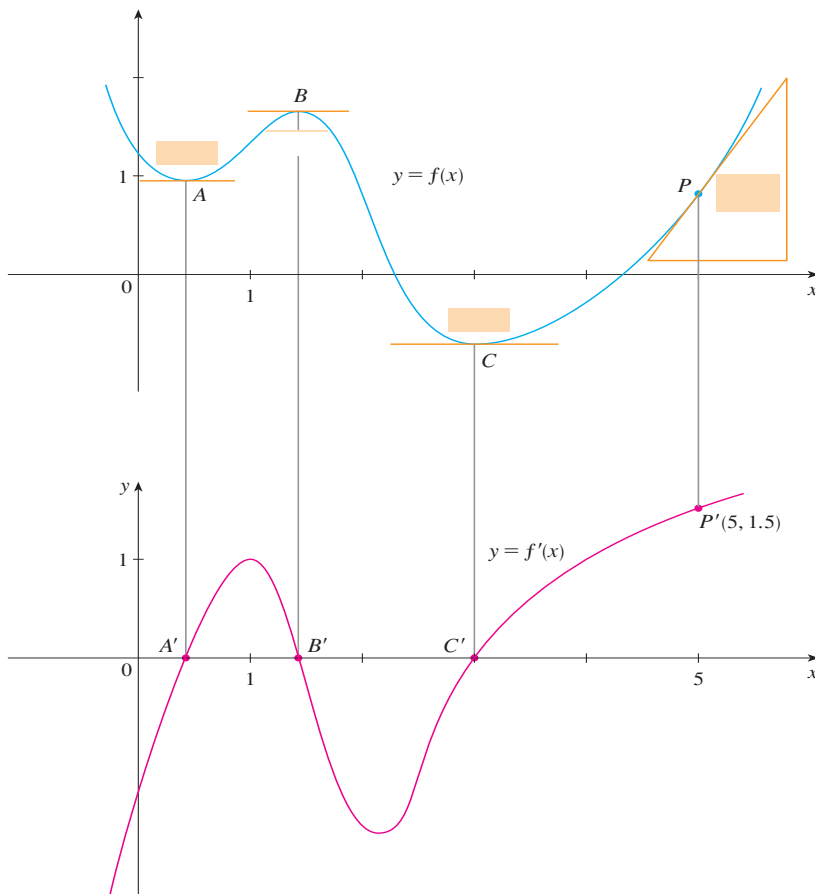
Here we change our point of view and let the number a vary. If we replace a in Equation 1 by a variable x , we obtain

$$\boxed{2} \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Given any number x for which this limit exists, we assign to x the number $f'(x)$. So we can regard f' as a new function, called the **derivative of f** and defined by Equation 2. We know that the value of f' at x , $f'(x)$, can be interpreted geometrically as the slope of the tangent line to the graph of f at the point $(x, f(x))$.

EXAMPLE 1 The graph of a function f is given in Figure 1. Use it to sketch the graph of the derivative f'

SOLUTION



EXAMPLE 2

- (a) If $f(x) = x^3 - x$, find a formula for $f'(x)$.
(b) Illustrate this formula by comparing the graphs of f and f' .

SOLUTION

EXAMPLE 3 If $f(x) = \sqrt{x}$, find the derivative of f . State the domain of f' .

SOLUTION

EXAMPLE 4 Find f' if $f(x) = \frac{1-x}{2+x}$.

SOLUTION

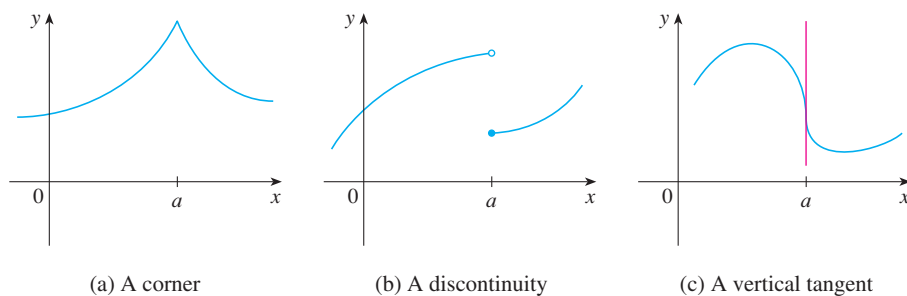
3 Definition A function f is **differentiable at a** if $f'(a)$ exists. It is **differentiable on an open interval (a, b)** [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

EXAMPLE 5 Where is the function $f(x) = |x|$ differentiable?

SOLUTION

4 Theorem If f is differentiable at a , then f is continuous at a .

How Can a Function Fail To Be Differentiable?



A function the curve has a **vertical tangent line** when $x = a$; that is, f is continuous at a and $\lim_{x \rightarrow a} |f'(x)| = \infty$

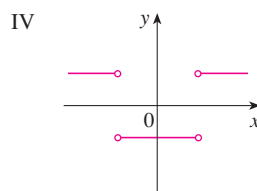
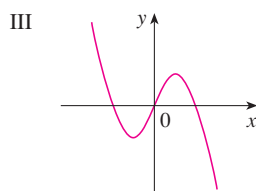
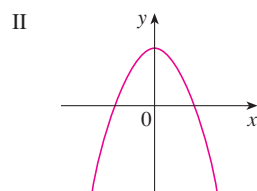
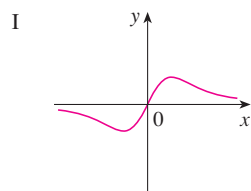
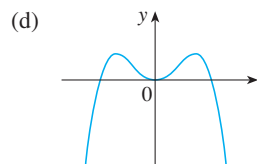
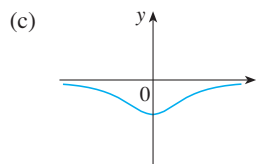
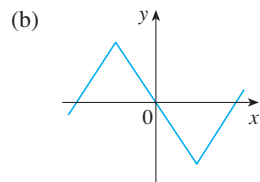
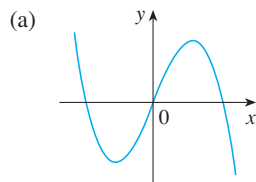
A function the curve has a **corner** when $x = a$; that is, f is continuous at a and

Higher Derivatives

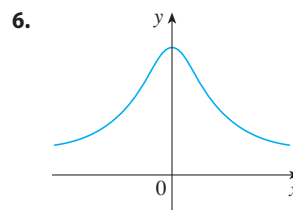
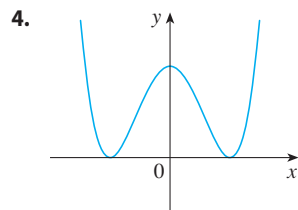
second derivative

$$\underbrace{\frac{d}{dx}}_{\text{derivative of}} \underbrace{\left(\frac{dy}{dx} \right)}_{\text{first derivative}} = \underbrace{\frac{d^2y}{dx^2}}_{\text{second derivative}}$$

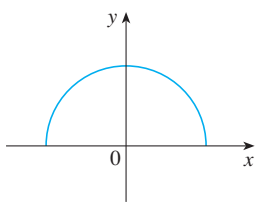
3. Match the graph of each function in (a)–(d) with the graph of its derivative in I–IV. Give reasons for your choices.



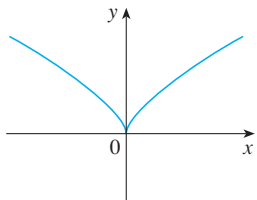
4–11 Trace or copy the graph of the given function f . (Assume that the axes have equal scales.) Then use the method of Example 1 to sketch the graph of f' below it.



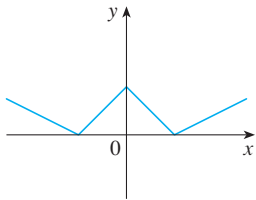
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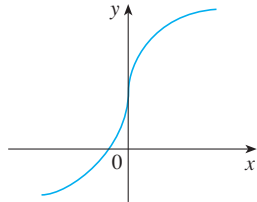
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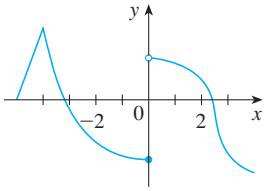


11.

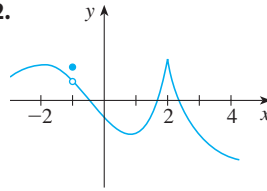


41–44 The graph of f is given. State, with reasons, the numbers at which f is *not* differentiable.

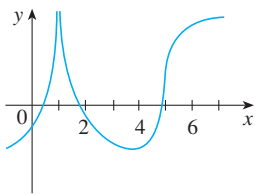
41.



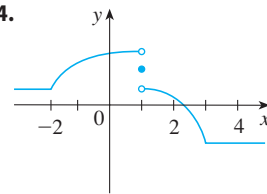
42.



43.



44.



57. Let $f(x) = \sqrt[3]{x}$.

- If $a \neq 0$, find $f'(a)$.
- Show that $f'(0)$ does not exist.
- Show that $y = \sqrt[3]{x}$ has a vertical tangent line at $(0, 0)$.

58. (a) If $g(x) = x^{2/3}$, show that $g'(0)$ does not exist.
(b) If $a \neq 0$, find $g'(a)$.
(c) Show that $y = x^{2/3}$ has a vertical tangent line at $(0, 0)$.
(d) Illustrate part (c) by graphing $y = x^{2/3}$.

-
61. (a) Sketch the graph of the function $f(x) = x|x|$.
(b) For what values of x is f differentiable?
(c) Find a formula for f' .

-
63. Recall that a function f is called *even* if $f(-x) = f(x)$ for all x in its domain and *odd* if $f(-x) = -f(x)$ for all such x . Prove each of the following.
(a) The derivative of an even function is an odd function.

-
- (b) The derivative of an odd function is an even function.